

Assignment - 01

Analogy and Mathematical Modeling of Boost and Buck-Boost Converters

Introduction

The boost and buck-boost converters are two basic non-isolated DC-DC power converter topologies. Both are built from the same four elements: an inductor (L), a controlled semiconductor switch (S), a diode (D) and an output capacitor (C) supplying a load resistance (R). What separates them is the way these elements are arranged between the source and the load, and that arrangement decides the voltage gain and the output polarity of

This assignment first presents the circuit configuration of each converter together with its simulation, then compares the two converters, derives a mathematical model of each from the two switching states, and finally applies the buck-boost model to a solar (photovoltaic) plant, where the converter is used as the interface between the solar panel and the load in a maximum power point tracking (MPPT) scheme.

1. Basic Circuit Configuration

1.1 Boost Converter

In the boost converter the inductor is connected in series with the input source, and the switch is connected across the circuit after the inductor. When the switch is closed the source charges the inductor. When the switch opens the inductor current is forced through the diode into the output capacitor and the load, so the inductor voltage adds to the source voltage. The output voltage is therefore always higher than the input voltage and has the same polarity. Because the inductor is at the input, the input current is continuous.

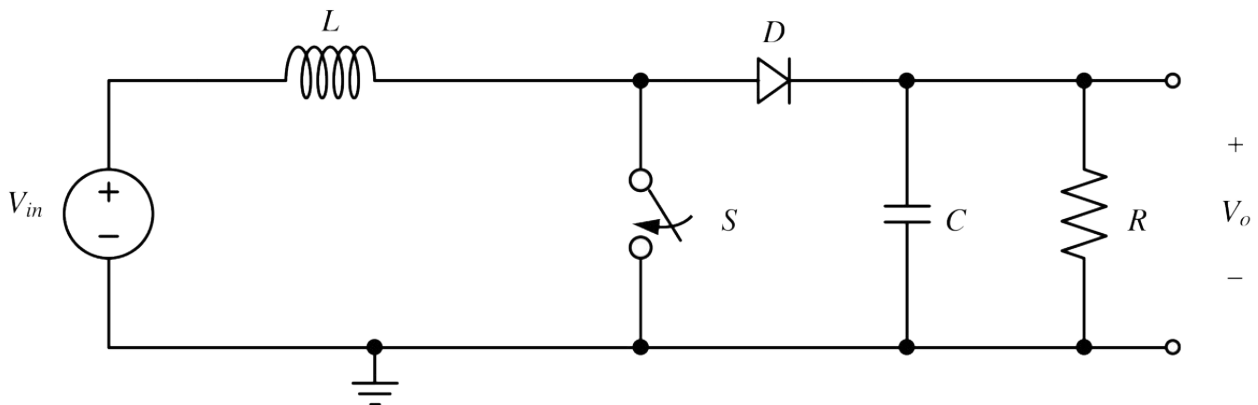


Figure 1: Circuit diagram of the boost converter

1.2 Buck-Boost Converter

In the buck-boost converter the switch is connected in series with the source and the inductor is connected across the circuit after the switch. When the switch is closed the whole input voltage appears across the inductor, and the diode is reverse biased. When the switch opens, the inductor current continues to flow through the diode into the output capacitor and the load, in the reverse direction with respect to the source. The output voltage is therefore of *inverted polarity*, and its magnitude may be smaller or larger than the input, depending on the duty cycle.

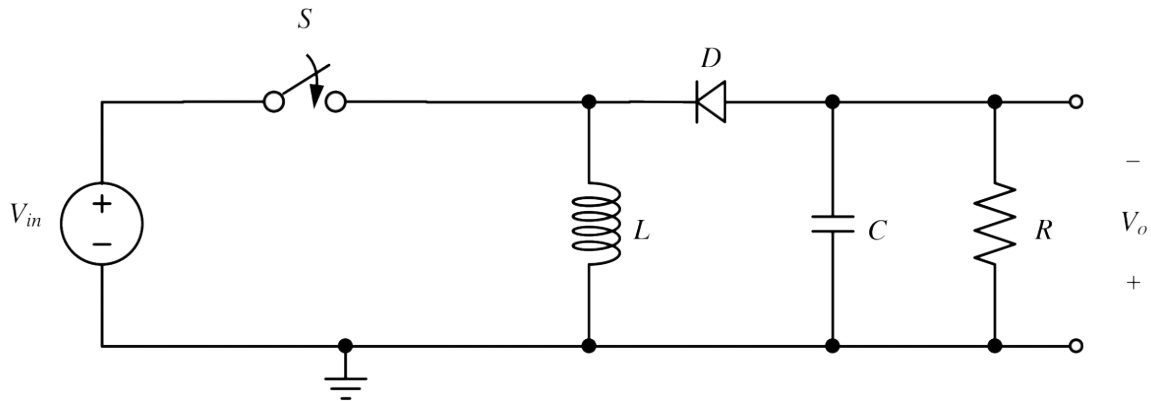


Figure 2: Circuit diagram of the buck-boost converter (inverting)

2. Circuit Simulation

Both converters were simulated in a circuit simulator with a 5 V DC source, a 10 mH inductor, a 10 μ F output capacitor and a 1 k Ω load. A MOSFET is used as the switch in both cases.

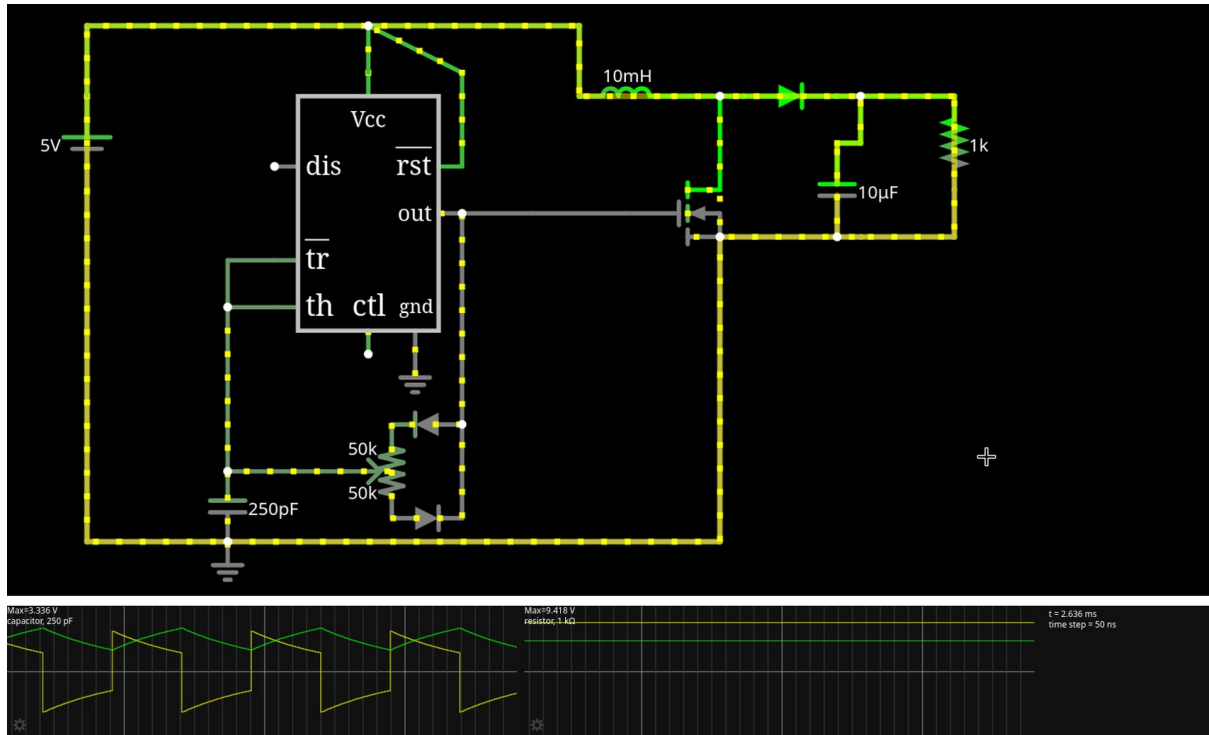


Figure 3: Boost converter simulation. The MOSFET gate is driven by a 555 timer in astable mode (upper part: circuit; lower part: timer capacitor waveform and load voltage)

In Figure 3 the load voltage settles at about 9.4 V from a 5 V source, a voltage gain of roughly 1.9. This confirms that the boost converter steps the voltage up while keeping the same polarity as the input.

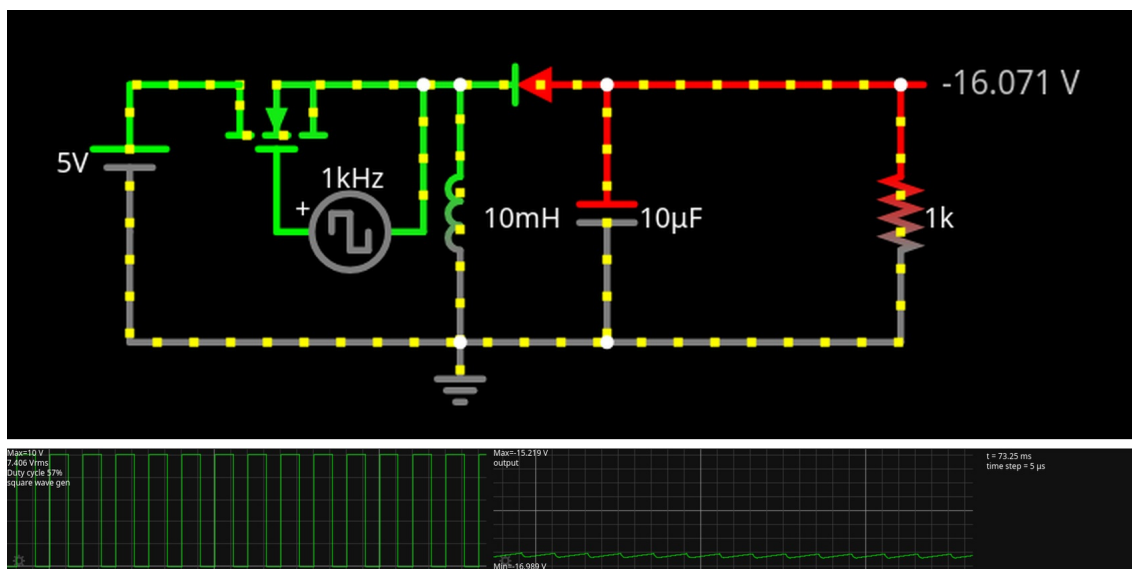


Figure 4: Buck-boost converter simulation. The MOSFET gate is driven by a 1 kHz square-wave generator (upper part: circuit; lower part: gate signal and output voltage)

In Figure 4 the output voltage is about -16 V (between -15.2 V and -17.0 V, the difference being the output ripple) from a +5 V source. This confirms the inverting nature of the buck-boost converter: a positive input produces a negative output with respect to the common

ground.

Note. The gain formulas derived in Sections 4 and 5 assume continuous conduction mode (CCM). With the light load and the small inductance used in this simulation, the inductor current falls to zero during part of each cycle (discontinuous conduction mode). In this mode the gain is higher than the CCM value; for example, at the 57% duty cycle shown by the generator, the CCM formula gives $|V_o| = 6.6$ V, which is lower than the 16 V observed. The CCM equations hold when the inductance is greater than the critical value $L_{\text{crit}} = R(1-D)^2/(2f)$.

3. Analogy Between Boost and Buck-Boost

The two converters are compared in Table 1.

Feature	Boost converter	Buck-boost converter
Components	L, S, D, C	L, S, D, C (same)
Switch position	Shunt (across the circuit)	Series with the source
Inductor position	Series with the source	Shunt (across the circuit)
Output polarity	Same as input (positive)	Inverted (negative)
Output magnitude	Always $\geq V_{\text{in}}$ (step-up only)	Smaller or larger than V_{in} (step-down and step-up)
Inductor in OFF state	Connected to the load in series with the source (input current is continuous)	Disconnected from the source, delivers its stored energy to the load only
Input current	Continuous	Pulsating
Voltage gain	$V_o/V_{\text{in}} = 1/(1-D)$	$ V_o /V_{\text{in}} = D/(1-D)$

Table 1: Comparison of boost and buck-boost converters

Key mathematical analogy. The gain of the buck-boost converter is conceptually the product of the gain of a buck converter and the gain of a boost converter operated with the same duty cycle:

$$\frac{V_o}{V_{\text{in}}} \Big|_{\text{buck}} \times \frac{V_o}{V_{\text{in}}} \Big|_{\text{boost}} = D \times \frac{1}{1-D} = \frac{D}{1-D} = \frac{|V_o|}{V_{\text{in}}} \Big|_{\text{buck-boost}} \quad (1)$$

So the buck-boost converter can be seen as a buck stage and a boost stage combined into one inductor, one switch and one diode. As a result it can step the voltage down when $D < 0.5$ and step it up when $D > 0.5$, whereas the boost converter can only step up (Figure 5). Both gains grow very rapidly as D approaches 1, so in practice the duty cycle is limited to a value below about 0.9 because of losses.

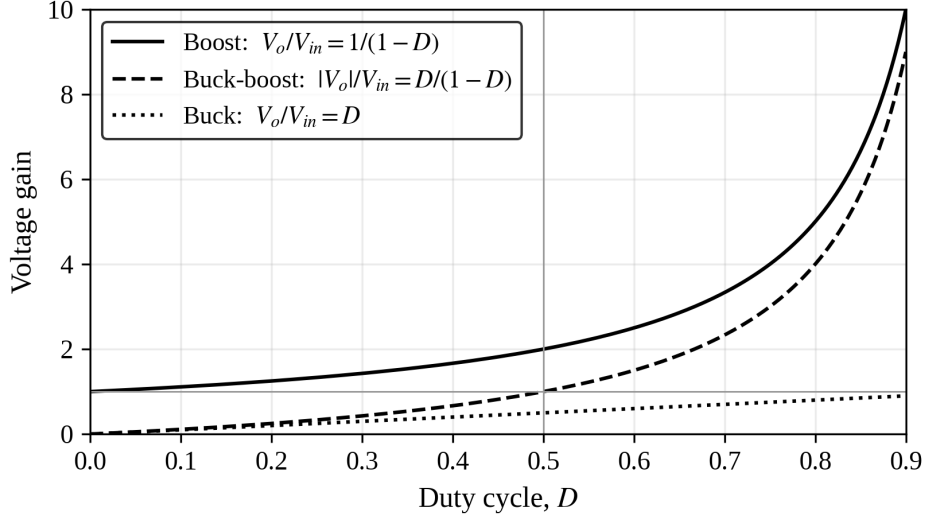


Figure 5: Ideal voltage gain of the boost, buck-boost and buck converters versus duty cycle

4. Mathematical Model of the Boost Converter

The converter is analysed over one switching period T with duty cycle D . The state variables are the inductor current i_L and the capacitor voltage v_C , and all components are assumed ideal.

Switch ON ($0 < t < DT$). The inductor is connected across the source and charges. The diode is reverse biased, so the capacitor alone supplies the load.

$$L \frac{di_L}{dt} = V_{in} \quad C \frac{dv_C}{dt} = -\frac{v_C}{R} \quad (2)$$

Switch OFF ($DT < t < T$). The diode conducts, and the inductor and the source together feed the capacitor and the load.

$$L \frac{di_L}{dt} = V_{in} - v_C \quad C \frac{dv_C}{dt} = i_L - \frac{v_C}{R} \quad (3)$$

Averaged state-space model. Weighting the ON-state equations by d and the OFF-state equations by $(1-d)$ and adding them gives

$$L \frac{di_L}{dt} = V_{in} - (1-d)v_C \quad (4)$$

$$C \frac{dv_C}{dt} = (1-d)i_L - \frac{v_C}{R} \quad (5)$$

Steady state. In steady state the average inductor voltage is zero (volt-second balance) and the average capacitor current is zero (charge balance). Setting the derivatives to zero, with V_o and I_L as the steady-state values:

$$V_{in}D + (V_{in} - V_o)(1-D) = 0 \Rightarrow \frac{V_o}{V_{in}} = \frac{1}{1-D} \quad (6)$$

$$(1-D)I_L = \frac{V_o}{R} \Rightarrow I_L = \frac{V_o}{R(1-D)} = \frac{V_{in}}{R(1-D)^2} \quad (7)$$

5. Mathematical Model of the Buck-Boost Converter

For the buck-boost converter, v_C denotes the magnitude of the capacitor voltage, so that the actual output voltage is $V_o = -v_C$.

Switch ON ($0 < t < DT$). The full input voltage is applied across the inductor. The diode blocks and the capacitor supplies the load.

$$L \frac{di_L}{dt} = V_{in} \quad C \frac{dv_C}{dt} = -\frac{v_C}{R} \quad (8)$$

Switch OFF ($DT < t < T$). The source is disconnected. The inductor current flows through the diode into the capacitor and the load, and the inductor voltage is equal to the output voltage.

$$L \frac{di_L}{dt} = -v_C \quad C \frac{dv_C}{dt} = i_L - \frac{v_C}{R} \quad (9)$$

Averaged state-space model.

$$L \frac{di_L}{dt} = dV_{in} - (1-d)v_C \quad (10)$$

$$C \frac{dv_C}{dt} = (1-d)i_L - \frac{v_C}{R} \quad (11)$$

Steady state. Volt-second balance on the inductor and charge balance on the capacitor give

$$V_{in}D - v_C(1-D) = 0 \Rightarrow \frac{|V_o|}{V_{in}} = \frac{D}{1-D} \quad (12)$$

$$(1-D)I_L = \frac{|V_o|}{R} \Rightarrow I_L = \frac{|V_o|}{R(1-D)} = \frac{DV_{in}}{R(1-D)^2} \quad (13)$$

For $D = 0.5$ the output magnitude equals the input, for $D < 0.5$ the converter steps down, and for $D > 0.5$ it steps up, with the output always negative with respect to the input ground.

6. Solar (PV) Plant Application

6.1 Why the Buck-Boost Converter is Preferred

The terminal voltage of a photovoltaic (PV) panel changes considerably with solar irradiance and cell temperature. Depending on the conditions, the panel voltage V_{pv} can be above or below the voltage required by the battery or DC bus. A buck converter can only step down and a boost converter can only step up, so neither can cover both situations. The buck-boost converter can, and for this reason it is preferred for the PV interface. (The inverted output polarity is not a problem for a DC bus or battery, as it only requires a reversed connection or an additional inverting stage.)

6.2 PV Source Model

A PV panel is modelled by the single-diode equivalent circuit. The panel current is

$$I_{pv} = I_{ph} - I_0 \left[\exp \left(\frac{q(V_{pv} + I_{pv}R_s)}{nkT} \right) - 1 \right] - \frac{V_{pv} + I_{pv}R_s}{R_{sh}} \quad (14)$$

where I_{ph} is the photo-generated current (proportional to irradiance), I_0 is the diode saturation current, R_s and R_{sh} are the series and shunt resistances, n is the diode ideality factor, q is the electron charge, k is the Boltzmann constant and T is the cell temperature. Figure 6 shows the resulting I-V and P-V characteristics, computed from this model for an example 36-cell panel ($I_{ph} = 3.5$ A, $R_s = 0.3$ Ω , $R_{sh} = 300$ Ω , $n = 1.3$, 25 °C). The power is maximum at a single point, the maximum power point (MPP), at which the panel voltage and current are V_{mpp} and I_{mpp} . This point corresponds to an optimum resistance $R_{mpp} = V_{mpp}/I_{mpp}$, which is about 6 Ω for the example panel.

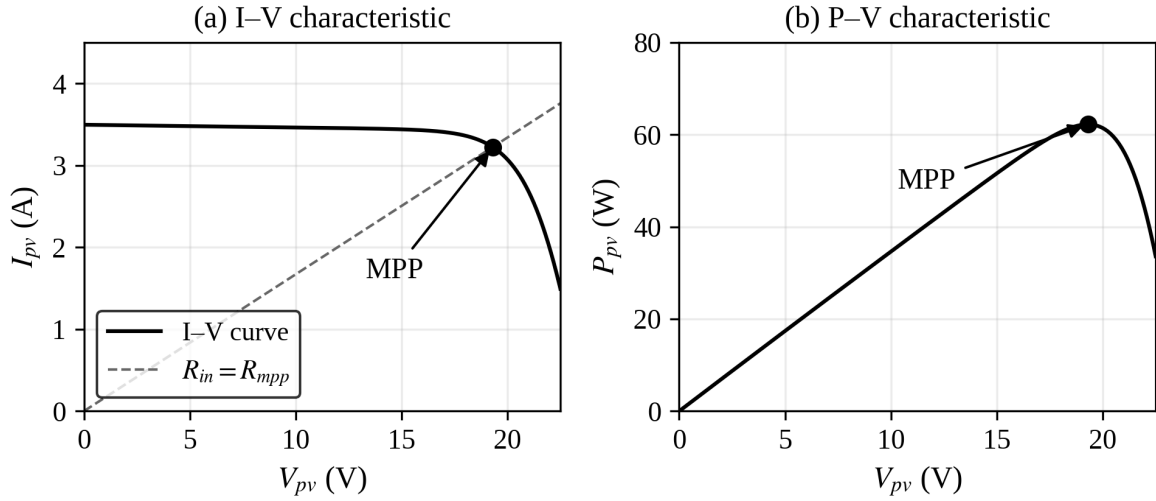


Figure 6: I-V and P-V characteristics of the example PV panel obtained from the single-diode model

6.3 Coupling the PV Panel with the Buck-Boost Converter

Assuming lossless conversion, the power taken from the panel is equal to the power delivered to the load:

$$V_{pv} I_{pv} = \frac{V_o^2}{R_{load}} \quad (15)$$

Using the buck-boost gain $V_o = [D/(1-D)]V_{pv}$, the resistance seen by the panel at the converter input is

$$R_{in} = \frac{V_{pv}}{I_{pv}} = R_{load} \left(\frac{1-D}{D} \right)^2 \quad (16)$$

This is the central equation of the solar-plant model. The panel does not see the actual load; it sees R_{in} , which is set by the duty cycle. An MPPT algorithm such as Perturb and Observe (P&O) or Incremental Conductance adjusts D continuously so that $R_{in} = R_{mpp}$, which forces the panel to operate at its maximum power point whatever the irradiance, temperature or load. The complete arrangement is shown in Figure 7.

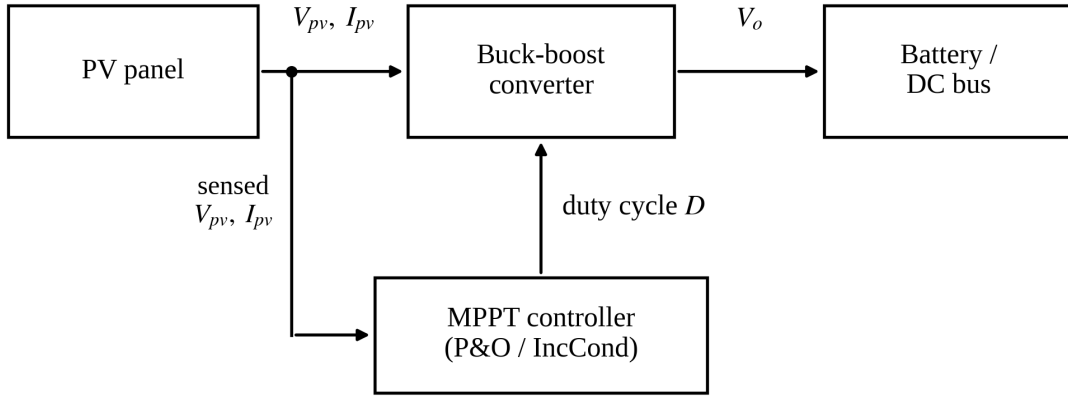


Figure 7: Block diagram of the PV system with buck-boost converter and MPPT controller

The converter topologies differ in the range of resistance they can present to the panel (Table 2).

Converter	Input resistance	Range of R_{in} ($0 < D < 1$)
Buck	R_{load}/D^2	R_{load} to infinity ($R_{in} \geq R_{load}$)
Boost	$R_{load}(1-D)^2$	0 to R_{load} ($R_{in} \leq R_{load}$)
Buck-boost	$R_{load}((1-D)/D)^2$	0 to infinity (any value)

Table 2: Input resistance presented to the PV panel by each converter

Only the buck-boost converter can present any value of R_{in} to the panel, so it can track the maximum power point for every combination of panel and load. A boost converter can only match a maximum power resistance smaller than the load resistance, and a buck converter only one larger than the load resistance.

6.4 Design Example

Consider the example panel of Figure 6 with $V_{mpp} = 19.3$ V, $I_{mpp} = 3.22$ A and $P_{mpp} = 62.2$ W (so $R_{mpp} = 5.99$ Ω), charging a 12 V battery bus. The equivalent load resistance is $R_{load} = 12^2/62.2 = 2.31$ Ω . From Equation (16) with $R_{in} = R_{mpp}$:

$$\frac{1-D}{D} = \sqrt{\frac{R_{mpp}}{R_{load}}} = \sqrt{\frac{5.99}{2.31}} = 1.61 \Rightarrow D = \frac{1}{1+1.61} = 0.383$$

The check gives $|V_o| = [0.383/(1-0.383)] \times 19.3 = 12.0$ V, as required. Here the converter steps the panel voltage down ($D < 0.5$). If the irradiance or the battery voltage changes so that the required output is higher than the panel voltage, the MPPT controller simply increases D above 0.5 and the same converter steps the voltage up, which a buck-only or boost-only converter could not do.

7. Conclusion

The boost and buck-boost converters are built from identical elements and differ only in the positions of the inductor and the switch. This changes the voltage gain from $1/(1-D)$ for the boost converter to $D/(1-D)$ for the buck-boost converter, and gives the buck-boost converter an inverted output that can be either smaller or larger than the input. The simulations confirmed a positive step-up output for the boost converter and a negative output for the buck-boost converter.

Both converters were modelled by writing the inductor and capacitor equations for the ON and OFF states, averaging them into a state-space model, and solving the steady state with volt-second and charge balance. In a solar plant the buck-boost converter is the natural choice, because its input resistance $R_{load}[(1-D)/D]^2$ can be set to any value by the duty cycle. An MPPT controller can therefore match it to the panel resistance R_{mpp} at all operating conditions, which a boost or buck converter alone cannot do.

References

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